

Grand Unification of Quantum Algorithms via Quantum Singular Value Transformation

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1. ABSTRACT & THEORETICAL FOUNDATION

We demonstrate that Shor factoring, Grover search, Hamiltonian simulation, HHL matrix inversion, and quantum walks are unified mathematical manifestations of polynomial transformations on block-encoded matrices.

2. CORE CONCEPTUAL INTUITION (PLAIN ENGLISH)

"A universal Swiss Army knife: QSVT showed that Shor's factoring, Grover's search, Hamiltonian simulation, and HHL are all just special cases of one master quantum formula."

3. Mathematical Formulation & Unitary Rule

$$U_{\Phi} = \prod_{k=1}^d \left(e^{i\phi_k(2\Pi - I)} U \right) = \begin{pmatrix} P(A) & \dots \end{pmatrix}$$

QSVT applies polynomial transformations to matrix singular values using alternating projector phase shifts.

5. Executable IBM Qiskit (Python) Code

```
# Block-encoding matrix A into unitary U = [[A, .], [. , .]]
# Sequence of phase angles Phi = [phi_0, phi_1, ... phi_d]
# transforms A -> P(A) via alternating reflections
print("QSVT unifies Grover, QPE, Hamiltonian simulation, and matrix inversion.")
```

6. Computational Complexity & Speedup

Classical Time:	Quantum Time:	Speedup Class:
$O(N^3)$ for full singular value decomposition	$O(\text{poly}(\log N))$ for block-encoding when A is sparse	Exponential in the degree of polynomial P

7. Key Applications & Future Research Directions

- Optimal Hamiltonian simulation matching Heisenberg uncertainty limits
- Unified matrix inversion and fractional matrix powers
- Quantum signal processing and quantum state filtering

Future Work:

Optimized phase angle synthesis, fractional query complexity, and general non-Hermitian operator processing.